ClipSAT : Digital SAT Math [ANSWER KEY]

Unit 0.01: Digital SAT Diagnostic & Adaptive Strategy Drill

Q1. If 2x-6 = 3x-5 , what is the value of x ?

[ ] A: -6 [ ] B: -8
[ ] C: -9 **[x] D: -7**

Solution: Distribute: 2x-12 = 3x-5 . -x = -5 + 12 Rightarrow x=-7 .

Q2. What is the sum of all solutions to the equation |2x-10| = 4 ?

[ ] A: 17 **[x] B: 10**
[ ] C: 7 [ ] D: 16

Solution: 2x-10=4 gives x=7 ; 2x-10=-4 gives x=3 . Sum =10 .

Q3. If dfrac{x + 8}{3} = 7 , what is the value of x ?

Solution: Multiply both sides by 3: x + 8 = 21 . Subtract 8 from both sides: x = 21 - 8 = 13 .

Q4. Suppose that 8x + 19 = 67 , what is the value of x ?

[x] A: 6 [ ] B: 7
[ ] C: 10.75 [ ] D: 48

Solution: Subtract 19 from both sides to get 8x = 48 , then divide by 8: x = 6 .

Q5. If 2x-6 = 3x-5 , what is the value of x ?

[ ] A: -6 [ ] B: -8
[ ] C: -9 **[x] D: -7**

Solution: Distribute: 2x-12 = 3x-5 . -x = -5 + 12 Rightarrow x=-7 .

Q6. begin{cases}x+6y=-43-x-y=13end{cases} What is the value of y ?

[x] A: -6 [ ] B: -9
[ ] C: -11 [ ] D: -2

Solution: Adding the equations eliminates x : 5y=-30 Rightarrow y=-6 . Then x=-7 .

Q7. begin{cases}6x-3y=-6-6x+5y=-2end{cases} What is the value of y ?

[x] A: -4 [ ] B: -7
[ ] C: 0 [ ] D: -6

Solution: Adding the equations eliminates x : 2y=-8 Rightarrow y=-4 . Then x=-3 .

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Q8. If 3x-2 = -3x-12 , what is the value of x ?

[ ] A: 1
[ ] B: 3
[x] C: -1
[ ] D: 5

Solution: Distribute: 3x-6 = -3x-12 . 6x = -12 + 6 Rightarrow x=-1 .